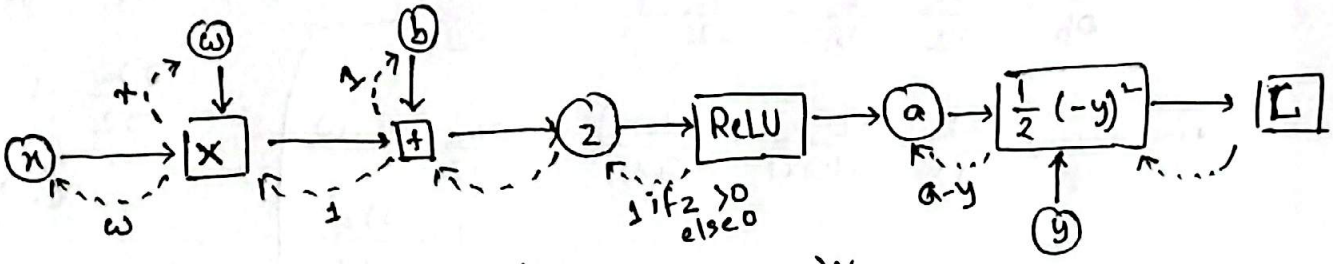
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CSE 715 | Quiz-2

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Question - 1



Local Jacobian $u \to v$ is defined as $J_{vu} = \frac{\partial v}{\partial u}$

Now calculate the derivatives

(i) from L to a : $L = \frac{1}{2}(a-y)^2$

$$\frac{\partial L}{\partial a} = 2 \cdot \frac{1}{2}(a-y) \cdot 1$$
$$= a - y$$

(ii) from a to z : $a = \max(0, z)$

$$\frac{\partial a}{\partial z} = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$

(iii) from z to addition (+) : $z = wx + b$

$$\frac{\partial z}{\partial (wx)} = 1 \quad , \quad \frac{\partial z}{\partial b} = 1$$

(iv) from addition (+) block to multiplication : $p = wx$

$$\frac{\partial p}{\partial w} = x \quad , \quad \frac{\partial p}{\partial x} = w$$

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derive $\frac{\partial L}{\partial a}, \frac{\partial L}{\partial z}, \frac{\partial L}{\partial w}, \frac{\partial L}{\partial x}$

(i) $\frac{\partial L}{\partial a} = (a-y)$

(ii) $\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} = (a-y) \cdot 1$ (if $z>0$) $= \begin{cases} a-y & \text{if } z>0 \\ 0 & \text{if } z<0 \end{cases}$

or, $(a-y) \cdot 0 = 0$ ($z<0$)

(iii) $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial (wx)} \cdot \frac{\partial (wx)}{\partial w}$

$= \frac{\partial L}{\partial z} \cdot 1 \cdot x$

$= x \cdot \frac{\partial L}{\partial z}$

(iv) $\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial b} = \frac{\partial L}{\partial z} \cdot 1 = \frac{\partial L}{\partial z}$

(v) $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial (wx)} \cdot \frac{\partial (wx)}{\partial x} = \frac{\partial L}{\partial z} \cdot 1 \cdot w$

$= w \cdot \frac{\partial L}{\partial z}$

Question-2

$x=2$, $w=3$, $b=-1$, $y=5$

Forward Pass:

$z = wx+b$

$= (3 \times 2) - 1$

$= 6-1$

$= 5$

$a = \max(0, z)$

$= \max(0, 5)$

$= 5$

$L = \frac{1}{2}(a-y)^2$

$= \frac{1}{2}(5-5)^2$

$= 0$

Backward pass:

(i) $\frac{\partial L}{\partial a} = a-y = 5-5 = 0$

(ii) $\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot 1 = 0 \cdot 1 = 0$ ($z=5$ which is >0, so $\frac{\partial a}{\partial z} = 1$)

(iii) $\frac{\partial L}{\partial w} = x \cdot \frac{\partial L}{\partial z} = 2 \cdot 0 = 0$

(iv) $\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} = 0$

(v) $\frac{\partial L}{\partial w} = w \cdot \frac{\partial L}{\partial z} = 3 \cdot 0 = 0$

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Question-3

When the input ~~is~~ of activition function is $z \leq 0$, the reverse path will change significantly.

(i) We will ~~encounter~~ encounter the gradient vanishing problem. As the local jacobian at ReLU nodes becomes 0.

(ii) We will experience dying ReLU problem. As $\frac{\partial L}{\partial a}$ is multiplied by jacobian of 0, the downstream gradient will become 0.

(iii) Upstream gradients: $\frac{\partial L}{\partial w}$, $\frac{\partial L}{\partial b}$, $\frac{\partial L}{\partial x}$ will evaluate to 0. As a result no signal will flow to weight (w) and bias (b). So it will not update during the gradient ~~flow~~ descent step.

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